Student Meal Planner

User Stories

An Analysis Document for the Grubs4Scrubs Project

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About This Document

These are the user stories for Grubs4Scrubs, grouped by feature area. Each story follows the standard format: "As a [user], I want [goal], so that [reason]." I also added acceptance criteria to the bigger stories so it's clear when they're actually done. The stories came out of the project analysis and the competitor research I did earlier in the semester.

Recipe Browsing

As a student, I want to browse a collection of budget-friendly recipes, so that I can find meals I can actually afford to make.

Acceptance criteria: - Recipes display as cards with an image, title, prep time, and estimated budget - Cards are clickable and lead to a full recipe view - The recipe list loads from the backend API

As a student, I want to filter recipes by category (breakfast, lunch, dinner, snack), so that I can find something specific without scrolling through everything.

Acceptance criteria: - Filter buttons appear above the recipe grid - Clicking a filter shows only recipes in that category - An "All" option resets the filter

As a student, I want to see a recipe's full details (ingredients, steps, nutrition, tips), so that I know exactly what I need and how to make it.

Acceptance criteria: - Clicking a recipe card opens a detailed view page - The page shows ingredients with a checkable list, step-by-step instructions, nutrition info, and a tip section - A hero image (or emoji fallback) appears at the top

Meal Planning

As a student, I want to plan my meals for each day of the week, so that I can organize what I'm eating and avoid buying random stuff.

Acceptance criteria: - A weekly view shows all seven days - Each day has slots for breakfast, lunch, and dinner - Users can see which slots are filled and which are empty

As a student, I want to see a weekly cost overview of my meal plan, so that I can stay within my budget.

Acceptance criteria: - The page shows total estimated cost across all planned meals - The count of planned meals is visible - Cost updates when meals are added or removed

Shopping List

As a student, I want a shopping list generated from my meal plan, so that I don't have to manually write down every ingredient.

Acceptance criteria: - The shopping list pulls ingredients from planned meals - Items can be checked off as purchased - The list groups items logically (not scattered randomly)

As a student, I want to manually add items to my shopping list, so that I can include things that aren't part of a recipe.

Acceptance criteria: - An "Add Item" form lets me type a name, quantity, and price - The new item appears in the list immediately after adding - The item is saved to the database through the API

As a student, I want to edit items on my shopping list, so that I can fix typos or update quantities without deleting and re-adding.

Acceptance criteria: - Each item has an edit button (pencil icon) - Clicking edit switches the row to inline input fields for name, quantity, and price - Saving the edit sends a PUT request to the API - Canceling the edit reverts the row to its original values

As a student, I want to delete items from my shopping list, so that I can remove things I no longer need.

Acceptance criteria: - Each item has a delete button (trash icon) - Clicking delete removes the item from the database through the API - The item disappears from the list without a full page reload

Recipe Management

As a student, I want to create new recipes, so that I can add my own meals to the collection.

Acceptance criteria: - A modal form collects all recipe fields (title, description, ingredients, instructions, budget, nutrition, etc.) - Submitting the form sends a POST request to the API - The new recipe appears in the recipe grid after creation

As a student, I want to edit a recipe I'm viewing, so that I can fix mistakes or update the instructions.

Acceptance criteria: - The recipe view page has an edit button (pencil icon) - Clicking it opens a modal pre-filled with the recipe's current data - Ingredients and instructions are displayed as editable text (one per line) - Saving sends a PUT request to the API and refreshes the displayed recipe

As a student, I want to delete a recipe, so that I can remove meals I don't want anymore.

Acceptance criteria: - The recipe view page has a delete button (trash icon) - Clicking delete removes the recipe through the API - The user is redirected back to the recipes grid after deletion

Dashboard

As a student, I want a dashboard that gives me an overview of my week, so that I can see everything at a glance without clicking through five pages.

Acceptance criteria: - Dashboard shows today's meals, weekly budget summary, and quick stats - Featured recipes appear for inspiration - A mini shopping list preview is visible - Navigation to other pages is accessible from the dashboard

User Accounts

As a student, I want to create an account with my email and password, so that my meal plans and preferences are saved.

Acceptance criteria: - Registration form accepts email, username, and password - Passwords are hashed before storage (BCrypt) - Duplicate emails are rejected with a clear error

As a student, I want to log in with Google, so that I don't have to remember another password.

Acceptance criteria: - A "Sign in with Google" button is available on the login page - Google OAuth returns user info that gets stored or matched to an existing account - GoogleId field is nullable (not every user uses Google)

As a student, I want to log in and stay logged in while I browse, so that I don't get kicked out every time I change pages.

Acceptance criteria: - Login returns a JWT token - The token is stored client-side and sent with API requests - Protected routes redirect to login if there's no valid token

Priority (MoSCoW)

Must have: Recipe browsing, recipe detail view, recipe CRUD (create, edit, delete), meal planner weekly view, user registration and login (backend complete, frontend UI built), dashboard overview, shopping list with full CRUD and inline editing.

Should have: Shopping list tied to user accounts, Google OAuth, budget tracking on the meal planner, recipe filtering by category (done), frontend auth wiring (login/signup forms connected to API).

Could have: Dietary preference filters, recipe rating system, sharing meal plans, recipe ownership enforcement (only creator can edit/delete).

Won't have (this semester): Social features, multi-user household support, grocery store price comparison.